Kubernetes Cluster Orchestration

Part I – Introduction

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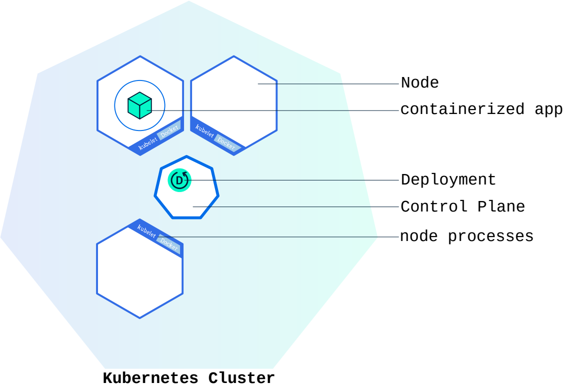
Kubernetes Concepts – Cluster

Kubernetes is a platform for managing a cluster of containers.

- Containerized app
  - Contains a single program and its dependencies
  - Depends on the OS kernel API
  - Can bind to network sockets and filesystem mounts
- Node
  - Handles a container instance
  - Interfaces with the cluster control

Kubernetes Concepts – Cluster



Kubernetes Concepts – Pod

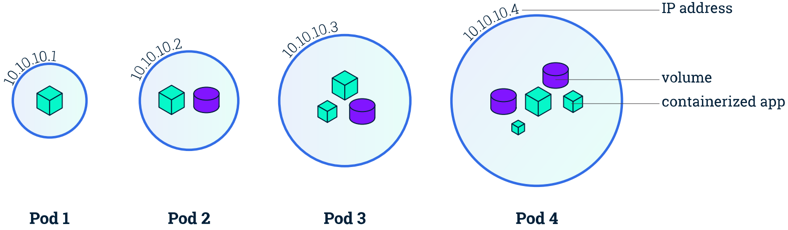
A Pod models an application built from smaller components.

- Shares IP sockets and storage volumes
- Has intra-pod networking

Example

Database ← application service ← web frontend

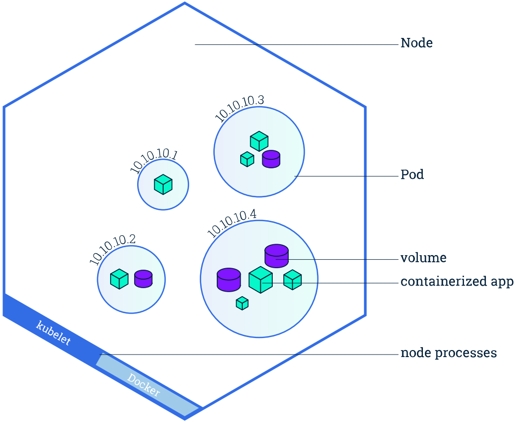
Kubernetes Concepts – Pod



Kubernetes Concepts – Node

A Node is a worker machine.

- Runs one or more Pods
- Managed by the controller through the *kubelet* process



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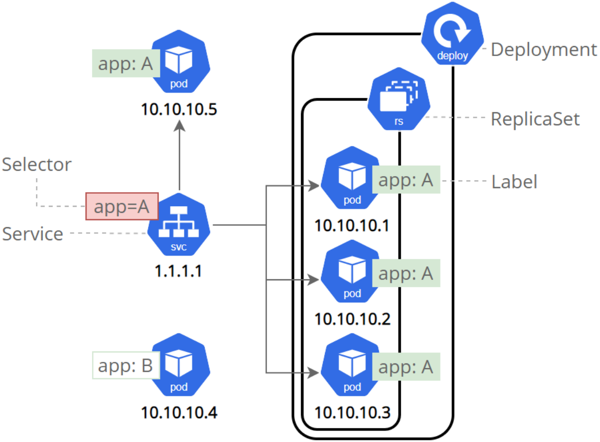
Application Deployment – Exposing an Application

Important concepts:

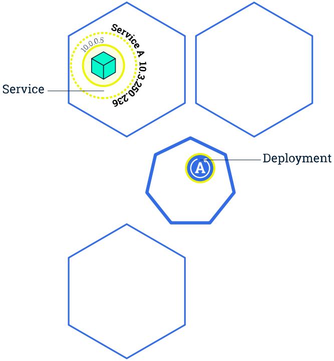
Replica Set	Identical, ephemeral worker pods
Service	Routes traffic across a set of pods
Label	E.g. production or test designation

Discovery and Routing is handled transparently.



Application Deployment – Scaling

- External interface is unchanged
- New pods are created
- Load balancing of nodes



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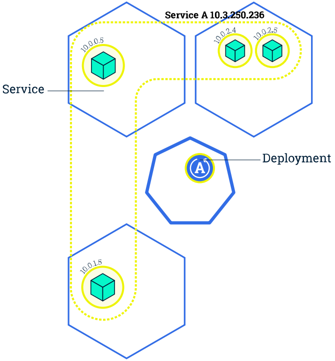


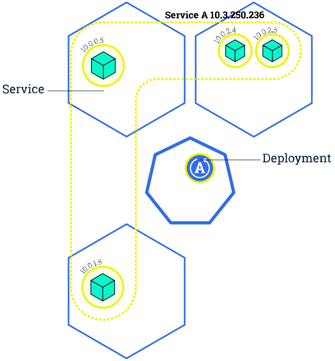
Application Deployment – Updates

Procedure:

1) Create new container image for new application
2) Deploy pods with new image
3) Retire old pods

Key benefit

Keeping some old pods online during transition provides zero downtime!



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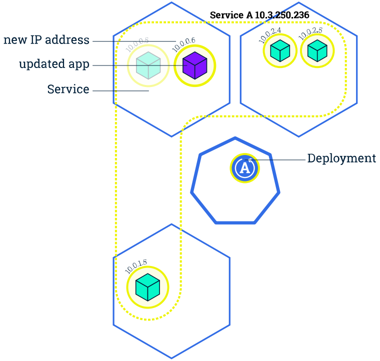

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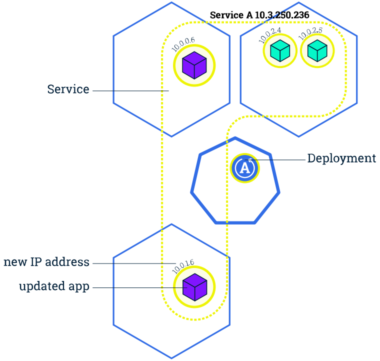

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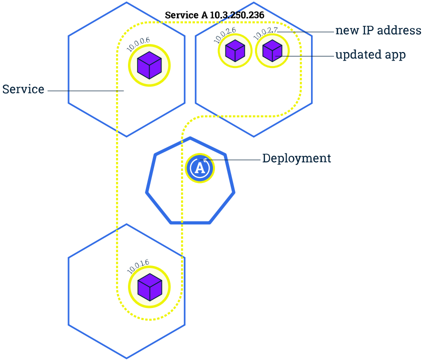

Appendix

Images and content taken from:
`https://kubernetes.io/docs/tutorials/kubernetes-basics/`